



Agenda

- ① Bit of human vision
- ② Channels
- ③ How CV started.
- ④ Convolution - Filters
- ⑤ Max Pooling



Archit Yesterday at 14:01

Hello @everyone

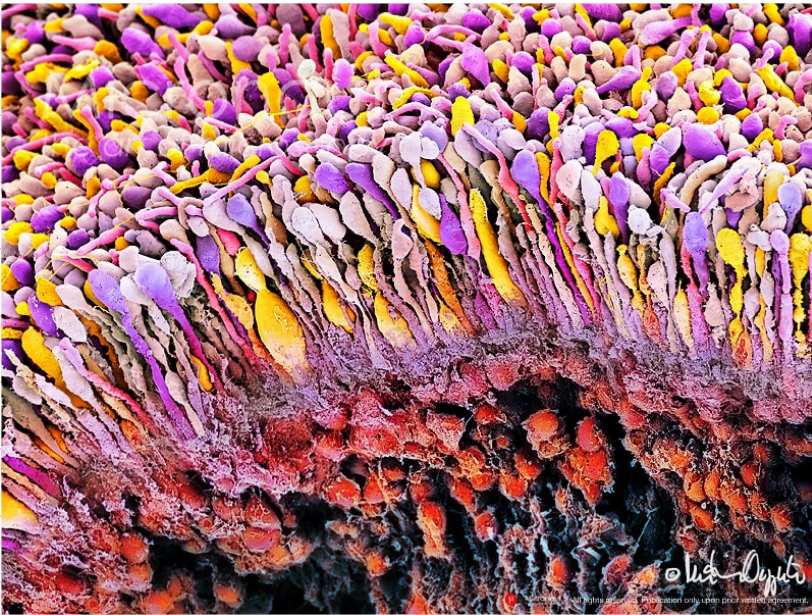
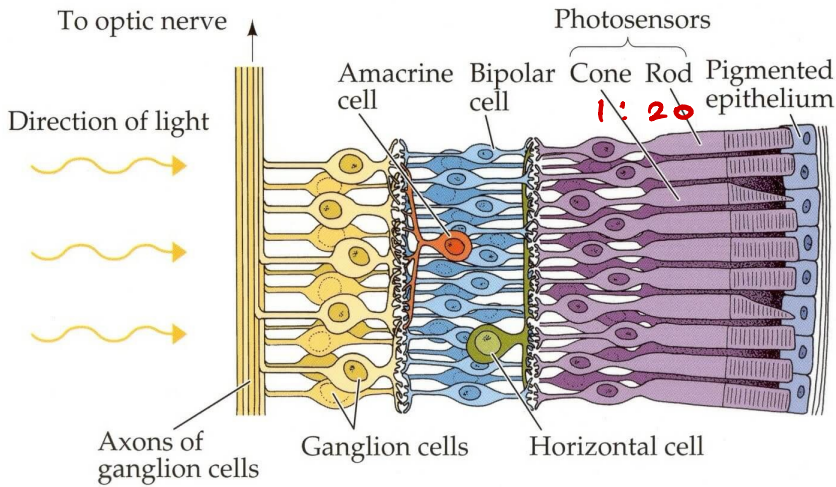
Please note that the IIT Roorkee instructor will be unavailable on **Tuesday, 28th July**, and **Thursday, 30th July**, due to an urgent meeting at IIT Roorkee.

As a result, **Shivam Sir** will be taking the classes on both days to ensure there is no disruption to your learning schedule.

Please join the sessions as per your regular class timings. Thank you for your understanding, and happy learning! (edited)



How Our Eyes Work

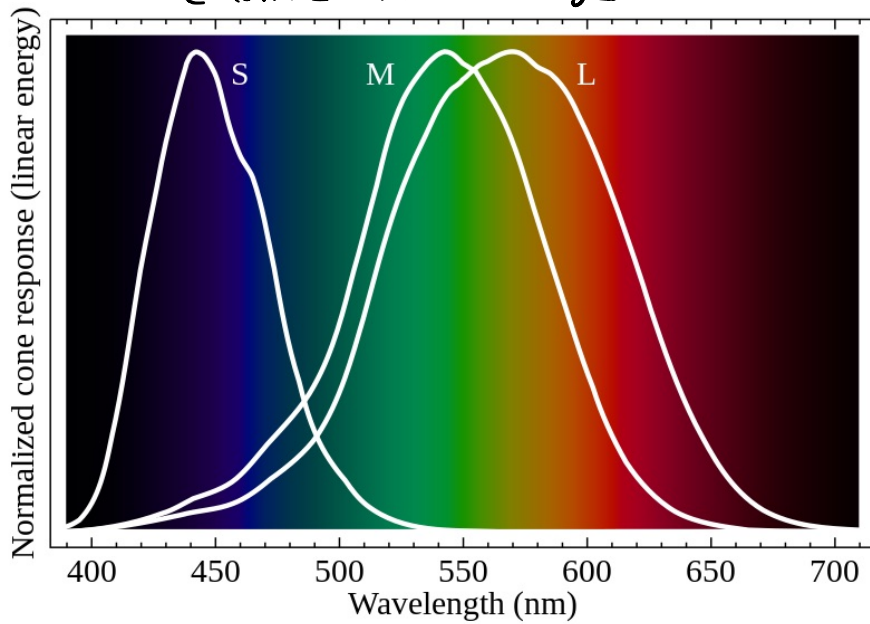


Channel → Container for similar information

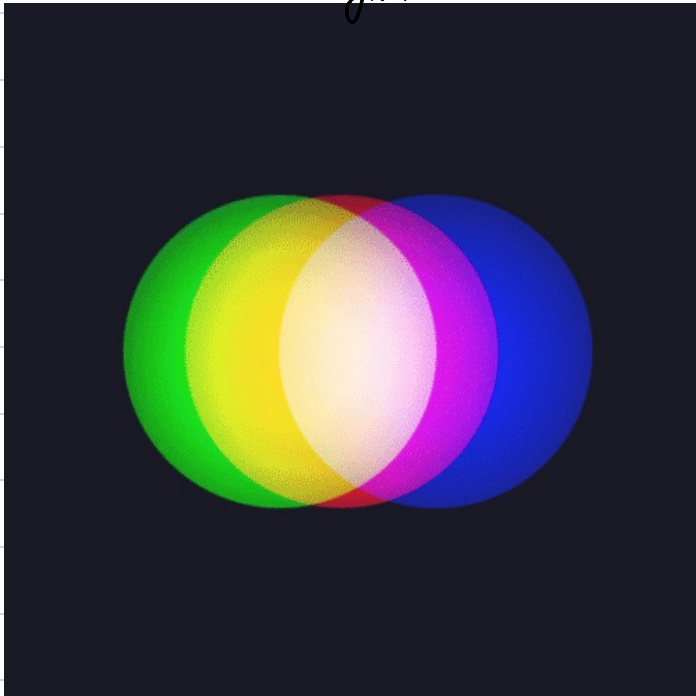


I said no gray m&m's, these are all gray.

Channels in our Eye



White is higher order channel



R	68	R	47	R	49	R	38	R
G	41	G	29	G	33	G	25	G
B	19	B	20	B	25	B	21	B
R	43	R	58	R	67	R	50	R
G	24	G	45	G	55	G	42	G
B	8	B	35	B	42	B	34	B
R	40	R	39	R	47	R	41	R
G	21	G	29	G	39	G	35	G
B	4	B	20	B	30	B	30	B
R	92	R	48	R	28	R	24	R
G	62	G	32	G	19	G	15	G
B	33	B	18	B	10	B	6	B
R	185	R	103	R	74	R	70	R

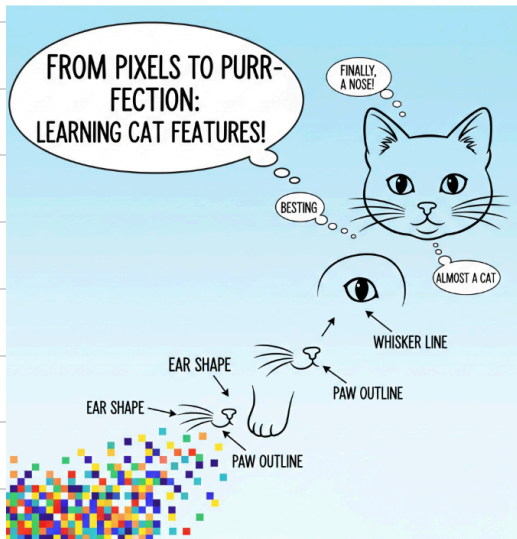
8bits

$$2^8 = 256$$

0 - 255

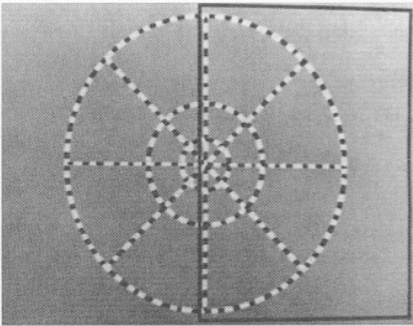
16 bits

0 - 65535

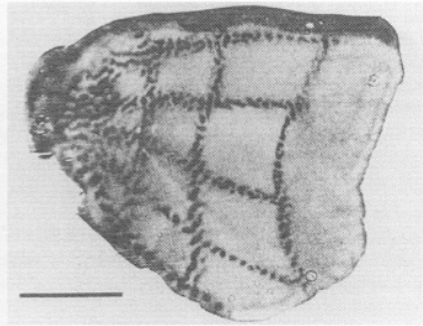


GRUESOME CAT Exp

gif-finder.com



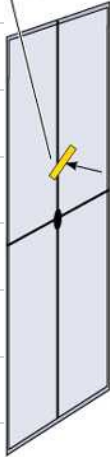
(a)



(b)

Experimental setup

Light bar stimulus
projected on screen



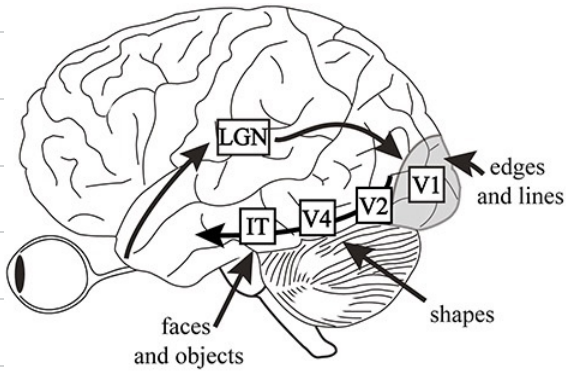
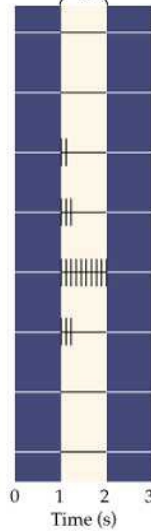
Recording from visual cortex

Record

Stimulus
orientation



Stimulus
presented



Receptive fields size

IT



V4



V2



V1



visual field

Features



faces



objects

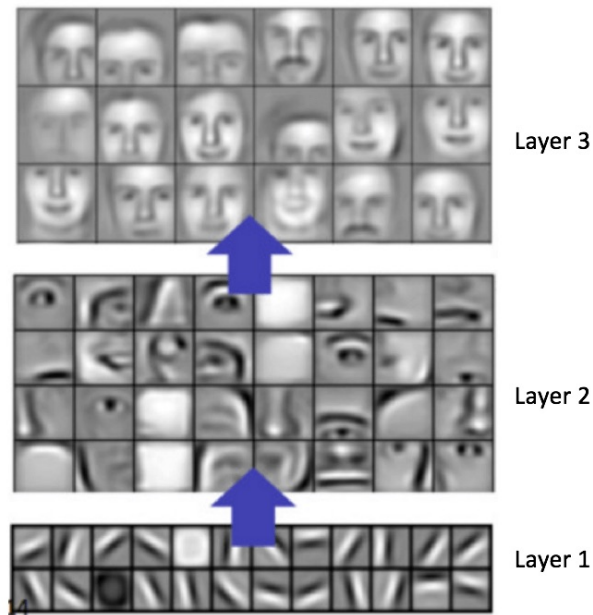
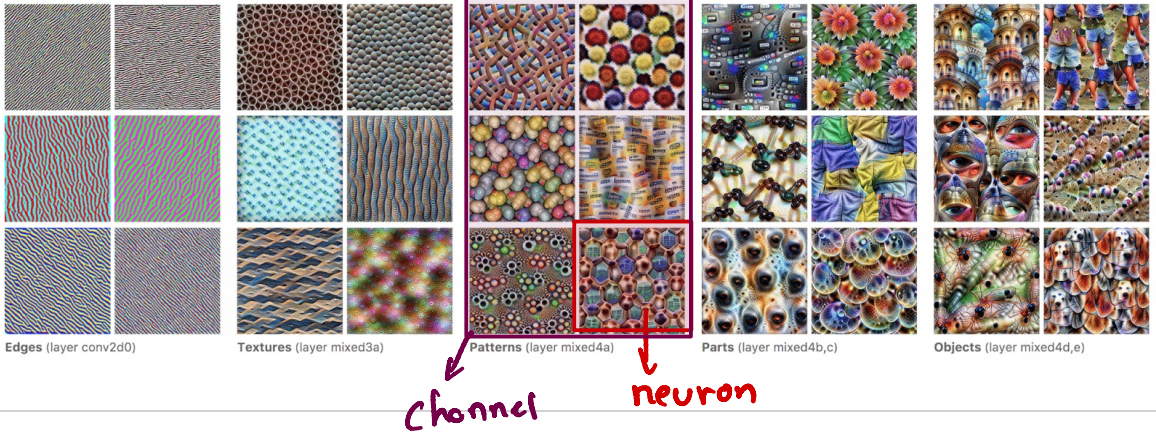


shapes



edges
and lines

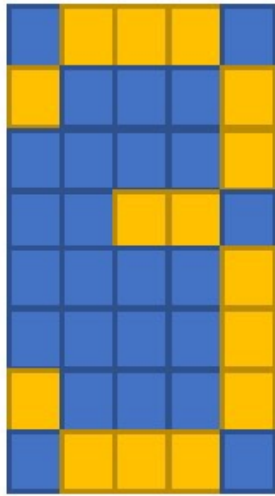
How channels look in each layer.



How to maintain spatial information in Neural Networks



What no. is this?? ↗↗



SOLUTION

CONVOLUTION

Kernel / Feature extractor / Filter

7	2	3	3	8
4	5	3	8	4
3	3	2	8	4
2	8	7	2	7
5	4	4	5	4

*

1	0	-1
1	0	-1
1	0	-1

=

6		

lots

$$7 \times 1 + 4 \times 1 + 3 \times 1 + 2 \times 0 + 5 \times 0 + 3 \times 0 + 3 \times -1 + 3 \times -1 + 2 \times -1 = 6$$

6	-9	-8
-3	-2	-3
-3	0	-2

How Convolution works??

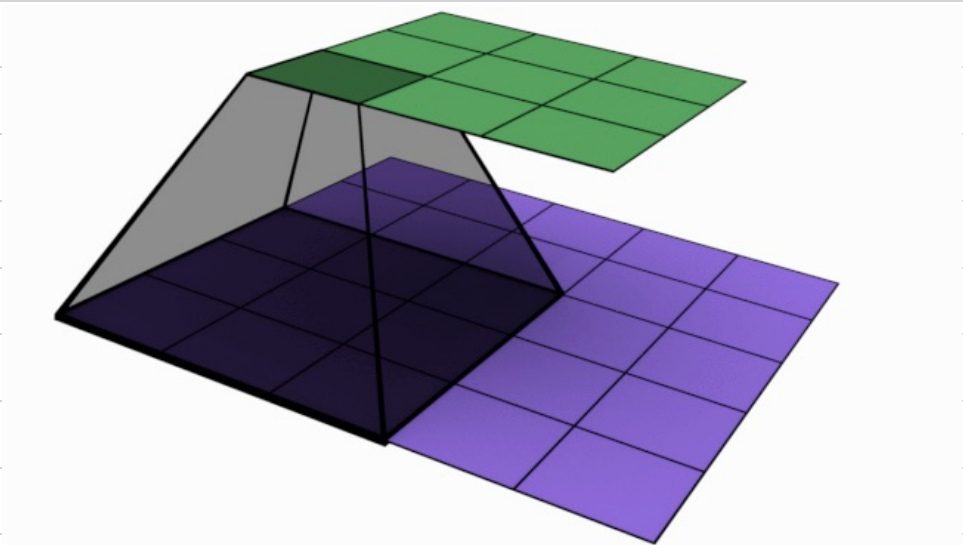
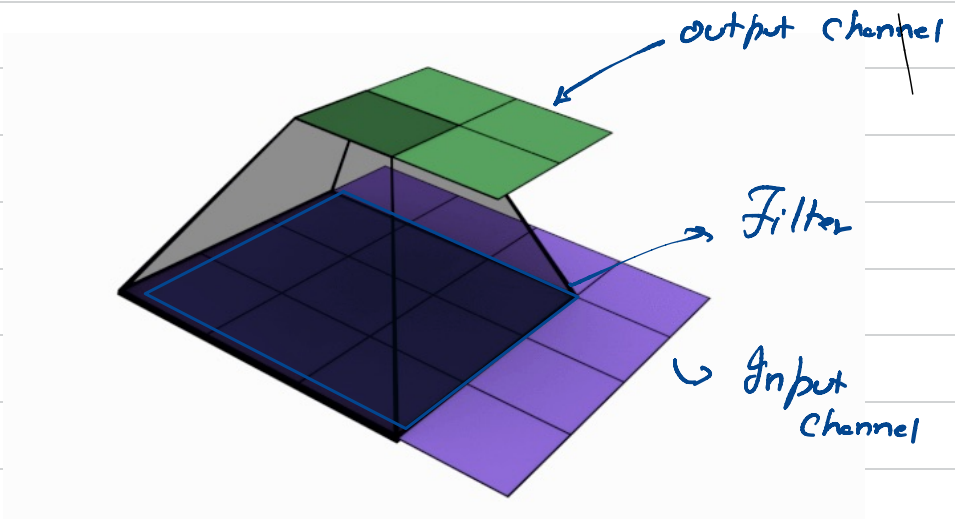
0.2	0.2	0.9	0.2	0.5
0.1	0.1	0.9	0.3	0.2
0.0	0.2	0.8	0.1	0.1
0.2	0.3	0.9	0.1	0.2
0.1	0.1	0.9	0.3	0.2

⊙

$$\begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$



$$\begin{bmatrix} -2.0 & 4.3 & -2.3 \\ -1.7 & 4.1 & -2.1 \\ -1.7 & 4.1 & -2.1 \end{bmatrix}$$



$$n_{out} = \left\lfloor \frac{n_{in} + 2p - k}{s} \right\rfloor + 1$$

n_{in} : number of input features

n_{out} : number of output features

k : convolution kernel size

p : convolution padding size

s : convolution stride size

① Input Image : 5×5

Kernel : 3×3

$p=0, s=1$

$$\left\lfloor \frac{5 + 2 \times 0 - 3}{1} \right\rfloor + 1 = 3 \times 3$$

② Input: 7×7

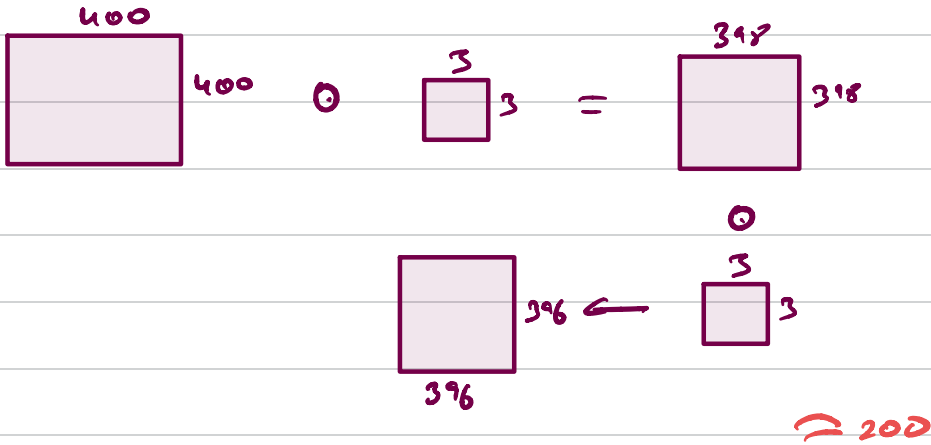
Kernel : 5×5

$p=0, s=1$

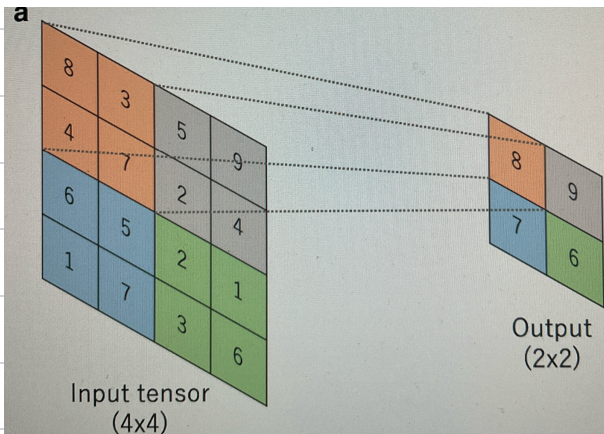
$$\left\lfloor \frac{7 + 2 \times 0 - 5}{1} \right\rfloor + 1 = 3 \times 3$$

→ Kernels act as Feature Extractors !!

→ How many layers would we need to move from 400×400 image to 1×1 !!



Max Pooling



0 1 0 0
0 1 0 0
0 1 0 0
0 1 0 0

1 0
1 0

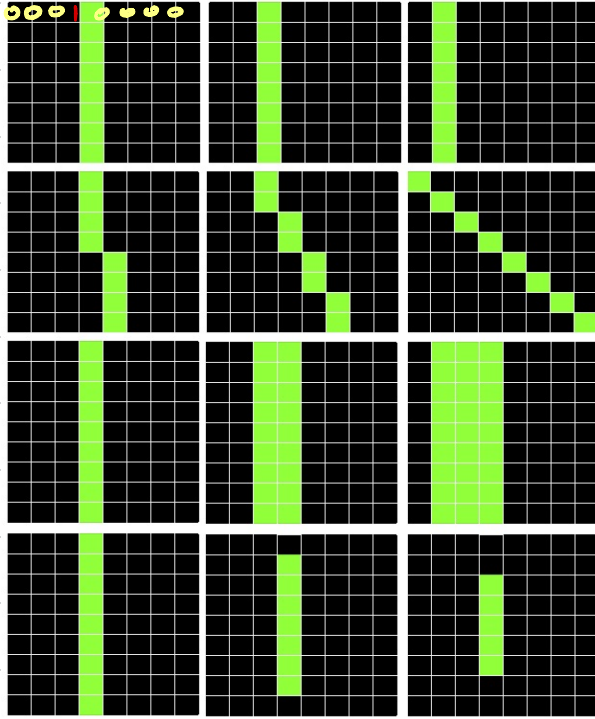
1 0
1 0

1 0 0 0
1 0 0 0
1 0 0 0
1 0 0 0

1 0
0 1

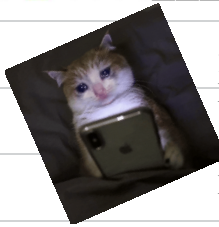


0 1 0 0
0 1 0 0
0 0 1 0
0 0 1 0



1 0
0 1
↑

1 0 0 0
0 1 0 0
0 0 1 0
0 0 0 1



Don't apply max pooling in the beginning part of the layer

Further, don't apply max pooling in the later part of the layers either

400 | 398 | 396 | 394 | 392 | 390 | MP (2x2)
195 | 193 | 191 | 189 | 187 | 185 | MP (2x2)
92 | 90 | 88 | 86 | 84 | 82 | MP (2x2)
41 | 39 | 37 | 35 | 33 | 31 | MP (2x2)
15 | 13 | 11 | 9 | 7 | 5 | 3 | 1

400 - 200 - 100 - 50 - 25 - 12 - 6 - 3 - 1

Number of ratings by lecture

Submitted ratings by lecture · 14-25 July 2026



The output shape when I convolve 400x400 with one 5x5 kernel with stride 1, and padding 0

The output shape when I convolve 400x400 with two 3x3 kernel with stride 1, and padding 0

Should we use 5x5 or 3x3?

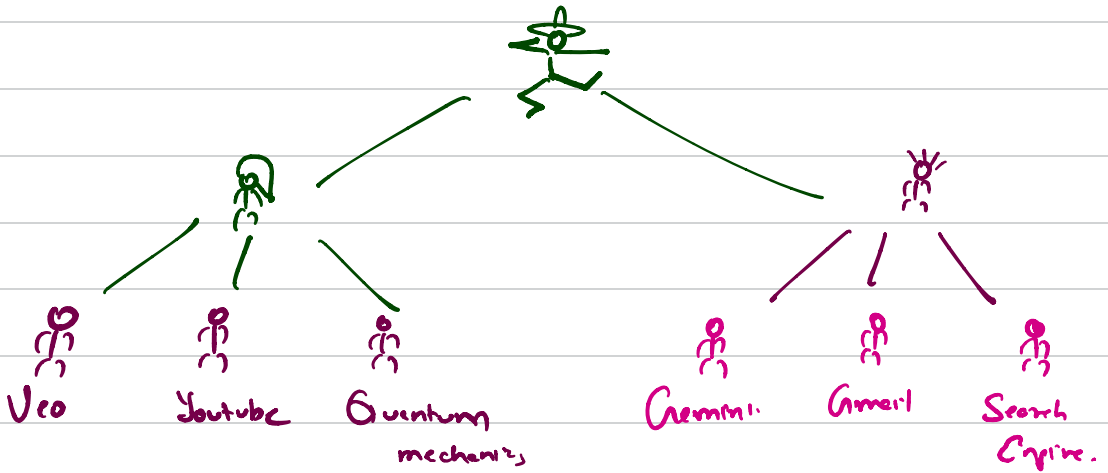
Poll results!

How many of you are learning AI/ML for the first time?

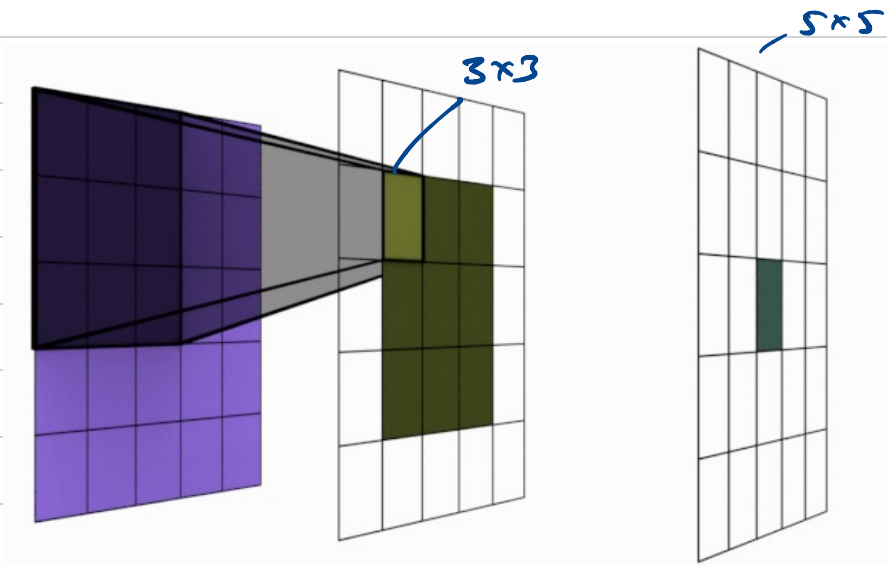
36 responses from 36 users



RECEPTIVE FIELD



Receptive field



What RF should you forget??